Zachary F. Mainen

Institution	Degree	Year	Field
Yale University, New Haven, CT, USA	B.A.	07/1990	Psychology
Yale University, New Haven, CT, USA	M.Sc.	07/1991	Psychology
University of California, San Diego, La Jolla, CA, USA	Ph.D.	07/1995	Neuroscience
Cold Spring Harbor Laboratory, Cold Spring Harbor, NY	Postdoc	09/1999	Neuroscience

Personal Statement

My research explores brain function and behavior using theory-driven experimental approaches. My laboratory pioneered the use of quantitative behavioral paradigms in rodents, combining those approaches with electrophysiological and optogenetic techniques to study neural coding and computation. I have a long-standing interest in the issue of how noise and uncertainty impact neural systems and behavior. A main theme in my work is the role of neuromodulators, especially serotonin, in neural circuit function and behavior. Through these themes, my ultimate aim is to bridge fundamental neuroscience to an understanding of human experience and well-being.

Positions, Scientific Appointments, and Honors

Positions

2007-present **Champalimaud Neuroscience Program**

2007-present Senior Group Leader

2007-2019 Founder and Director

Champalimaud Centre for the Unknown, Lisbon, Portugal

1999-2007 **Cold Spring Harbor Laboratory**

1999-2003 Assistant Professor

2003-2007 Associate Professor

2008-2010 Adjunct Associate Professor

Cold Spring Harbor, NY, USA

2008-2010 Janelia Farm Research Campus, Howard Hughes Medical Institute

Visiting Scientist

Ashburn, VA, USA

2002-2007 Department of Neurobiology & Behavior, Stony Brook University

Affiliated Associate Professor

Stony Brook, NY, USA

Scientific appointments

2016-present **International Brain Laboratory**, Founder and Executive Board member

2005-2024 Computational and Systems Neuroscience (Cosyne), Executive Board member

2011-present Cognitive and Computational Neuroscience Summer School, China

2011-2018 Co-Director

2023-present Advisory Board

2014-2017 CAJAL Advanced Neuroscience Training Program, Co-Director

2007-2011 International Neuroscience Doctoral Program, Champalimaud Foundation, Founding Director

2011, 2023 Champalimaud Neuroscience Symposium, Scientific Committee

Honors

2010-present European Molecular Biology Organization (EMBO) Elected Member

2018 Special Lecture, Society for Neuroscience Annual Meeting

2015-2020 European Research Council (ERC), Advanced Investigator Grant

2010-2015 European Research Council (ERC), Advanced Investigator Grant

2005 NARSAD, Lattner Foundation Investigator

2001-2005 Searle Scholar Award

1997-2002 Burroughs-Wellcome Career Development Award

1995-1997 National Institutes of Health NRSA Postdoctoral Fellowship

1991-1995 Howard Hughes Medical Institute Predoctoral Fellowship

Review boards

2021-present Max Planck Institute for Brain Research, Frankfurt, Scientific Advisory Board member

2016-present Current Opinion in Neurobiology, Editorial Board Member

2015-present Neural Computation, Editorial Board Member

2014-present Current Biology, Editorial Board Member

2019-2020 Review Commons, Advisory Board Member

2018-2020 Scientific Advisory Board, COMPASS Pathways Limited

2013-2018 NOVA Lisbon University, Portugal, Faculty of Sciences Council

2013-2015 European Research Council, Expert Panel Member

2010-2016 Nature Communications, Editorial Advisory Panel

2007-2014 Frontiers in Neuroscience, Associate Editor

2004-2005 National Institutes of Health (USA), Ad-hoc reviewer

Contributions to Science

1. Elucidation of the computational properties of the mammalian serotonin system

The serotonin system is one of the most important targets of psychoactive drugs and a key regulator of many behavioral functions, but its role in the nervous system has long been considered enigmatic. Starting with electrophysiological experiments and then incorporating advanced optogenetic approaches, my laboratory has pioneered the application of systems and computational methods to elucidating the function of serotonin. We received two major research grants from the European Research Council (over €5M total) to fund over 10 years of research. The central finding of our work is that serotonin neurons fire phasically in response to a diverse array of events which we propose comprise a latent state prediction error (Ranade et al., 2009; Matias et al., 2017; Cazettes et al., 2021). This signal in turn rapidly affects brain states and learning processes, contributing to cognitive flexibility and the restructuring of brain representations (Correia et al., 2017; Matias et al., 2017). Thus, we believe that serotonin plays a key role in orchestrating self-supervised learning, the foundation on which brain models of the organism and its environment are constructed. Recently, we showed that serotonin activity correlates with investigatory facial movements (Cazettes et al., 2025), and that optogenetic activation reconfigures brain-wide neural dynamics in a subspace orthogonal to ongoing task computations (Meijer et al., 2025). A central focus of our research is testing these ideas in diverse behavioral settings, using serotonin manipulations and recordings in combination with large-scale neural ensemble recordings from the cortex and other brain areas. These ideas have implications for understanding the function of serotonin-targeting drugs, including antidepressants and psychedelics.

Ranade SP, Mainen ZF (2009) Transient firing of dorsal raphe neurons encodes diverse and specific sensory, motor, and reward events. J Neurophysiol 102, 3026-37.

Matias S, Lottem E, Dugué GP, Mainen ZF (2017) Activity patterns of serotonin neurons underlying cognitive flexibility. eLife 6:e20552.

Correia PA, Lottem E, Banerjee D, Machado AS, Carey MR, Mainen ZF (2017) Transient inhibition and long-term facilitation of locomotion by phasic optogenetic activation of serotonin neurons. eLife 6:e20975.

Cazettes F, Reato D, Morais JP, Renart A, Mainen ZF (2021) Phasic activation of dorsal raphe serotonergic neurons increases pupil-linked arousal. Current Biology 31, 192–197.

Meijer GT, et al., Mainen ZF (2025) Serotonin modulates neural dynamics in a subspace orthogonal to the choice space. bioRxiv 10.1101/2025.08.01.668048.

2. Neural computation of decision confidence and uncertainty

The ability to optimally incorporate uncertainty into predictions is central to the notion of a predictive brain. Humans have a sense of confidence in decisions and other judgments, which has been considered a hallmark of metacognition. Such abilities were once generally thought to be exclusive to primates, and the computational mechanisms underlying a neural decision confidence signal were unknown. Work in my laboratory (Kepecs et al., 2008; Lak et al., 2014) made three major contributions to this area. First, we demonstrated that rats and mice can act optimally on decision confidence, extending this ability from primates to rodents. We posited that confidence estimation is a core feature of neural systems, and subsequent work supported this notion. Second, we were the first to record neural correlates of decision confidence in any species, helping to understand the neural underpinnings of these functions. Third, we proposed a theoretical account of the computations underlying decision confidence and its role in Bayesian decision-making (Kepecs and Mainen, 2012; Drugowitsch et al., 2019). Together, this work made foundational contributions to this area of research.

Kepecs A, Uchida N, Zariwala HA, Mainen ZF (2008) Neural correlates, computation and behavioral impact of decision confidence. Nature 455, 227-31.

Lak A, Costa GM, Romberg E, Koulakov AA, Mainen ZF, Kepecs A (2014) Orbitofrontal cortex is required for optimal waiting based on decision confidence. Neuron 84, 190-201.

Kepecs A, Mainen ZF (2012) A computational framework for the study of confidence in humans and animals. Phil Trans R Soc B 367, 1322–1337.

Drugowitsch J, Mendonça AG, Mainen ZF, Pouget A (2019) Learning optimal decisions with confidence. PNAS 116, 24872-24880.

3. Quantitative behavioral paradigms in rodents

Our work on decision confidence capitalized on experimental approaches for applying rigorous psychophysical methods in rodents that my lab helped to pioneer. When I launched my laboratory, awake behaving primate work was the gold standard in systems neuroscience. This work rested in large part on extremely rigorous behavioral training and testing, which allowed researchers to record hundreds or thousands of trials in a single session, repeated over periods of months. We took on the challenge of adapting these methods to rodents—first rats and later mice. The first major paper from my laboratory (Uchida and Mainen, 2003) is one of the seminal advances in the development of these approaches and played a pivotal role in sparking the explosion of rodent systems and computational neuroscience. Our work, together with collaborators Carlos Brody and Anthony Zador, showed that primate-level behavior could be obtained in rodents in a high-throughput manner. By combining rodent behavior with high-density neural ensemble recordings and optogenetic techniques for manipulating neural activity, the ”rodent systems neuroscience” approach has had a major impact on the practice of systems neuroscience. My lab has continued to pioneer novel behavioral paradigms in rodents, pushing the frontier in rats (Poo et al., 2022), freely behaving mice (Murakami et al., 2014), and head-fixed mice (Cazettes et al., 2023, 2025).

Uchida N, Mainen ZF (2003) Speed and accuracy of olfactory discrimination in the rat. Nature Neurosci 6, 1224-9.

Cazettes F, Mazzucato L, Murakami M, Morais JP, Augusto E, Renart A, Mainen ZF (2023) A reservoir of foraging decision variables in the mouse brain. Nature Neuroscience 26, 840-849.

Poo C, Agarwal G, Bonacchi N, Mainen ZF (2022) Spatial maps in olfactory cortex during olfactory navigation. Nature 601, 595-599.

Murakami M, Vicente MI, Costa GM, Mainen ZF (2014) Neural antecedents of self-initiated actions in secondary motor cortex. Nat Neurosci 17, 1574-82.

Cazettes F, Reato D, Augusto E, et al., Mainen ZF (2025) Facial expressions in mice reveal latent cognitive variables and their neural correlates. Nature Neuroscience 28, 2310-2318.

4. Electrophysiology and modeling of neocortical neurons and synapses

My training in neuroscience was in single-cell and single-synapse biophysics. I was fortunate to work with three extremely talented mentors: Terry Sejnowski, Roberto Malinow, and Karel Svoboda, from whom I learned a rich combination of computational modeling, biophysics, electrophysiology, and pharmacology. As a Ph.D. student, I focused on the nature of uncertainty or noise in the generation of action potentials by neocortical neurons. At the time, it was commonly believed that neurons could be considered to emit stochastic spike trains, meaning that the timing of individual spikes was unpredictable and meaningless. I showed using a relatively simple experiment, which has since been widely replicated, that if one injected a realistic pattern of current into a neuron, it would result in a highly reliable spike train. This paper (Mainen and Sejnowski, 1995) is considered a classic and among the most highly cited experimental results in neuroscience. It provided major support for the significance of the detailed timing of action potentials, a topic that continues to be actively investigated. The genesis of this paper arose from combining single-cell recording with detailed biophysical modeling. A second paper from this line of work (Mainen and Sejnowski, 1996), also part of my Ph.D. thesis, showed that the pattern of action potentials in neocortical neurons was highly influenced by the dendritic structure of the neuron, a fact that had never before been appreciated. This work was also influential, as it linked morphology and function in a novel way. As a postdoc, I went on to focus on synaptic function, investigating both the fundamental features of long-term potentiation and the biophysics of single synapses (Mainen et al., 1999). I consider that these studies, besides their impact, attest to my career-long track record of exploring different subfields of neuroscience, mastering and improving state-of-the-art techniques to yield important conceptual insights.

Mainen ZF, Malinow R, Svoboda K (1999) Synaptic calcium transients in single spines indicate that NMDA receptors are not saturated. Nature 399, 151-5.

Mainen ZF, Sejnowski TJ (1996) Influence of dendritic structure on firing pattern in model neocortical neurons. Nature 382, 363-6.

Mainen ZF, Sejnowski TJ (1995) Reliability of spike timing in neocortical neurons. Science 268, 1503-6.

5. Large-scale neuroscience collaborations to tackle brain-wide functions

Understanding brain function requires tools of unprecedented specificity and scale, but the brain's complexity means that effectively harnessing these tools is beyond the reach of single laboratories. As a result, new methods have mostly been applied piecemeal to study neural activity in single regions, and approaches vary across laboratories, making comparison and reproducibility difficult. Together with Michael Hausser and Alex Pouget, I founded the International Brain Laboratory (IBL), which was generously supported by the Simons and Wellcome Foundations. The IBL combined the efforts of 21 labs to understand the neural mechanisms of decision-making in mice: theorists and experimentalists work closely together, using standardized data processing pipelines and cloud-based data sharing. The IBL's flagship achievement was the first brain-wide map of neural activity during decision-making, recording from over 600,000 neurons across 279 brain areas (IBL, 2025). This work revealed that decision-related signals are distributed across nearly all recorded regions, with prior expectations encoded in 20–30% of areas from sensory to frontal cortex (Findling et al., 2025). We also demonstrated that electrophysiological measurements can be reproduced across laboratories when protocols are standardized (IBL, 2023). By developing strategies for data sharing, tight collaboration, and reproducibility, the IBL has helped pave the way for a new generation of large-scale neuroscience collaborations.

International Brain Laboratory et al. (2023) A modular architecture for organizing, processing and sharing neurophysiology data. Nat Methods, https://doi.org/10.1038/s41592-022-01742-6

International Brain Laboratory (2025) A brain-wide map of neural activity during complex behaviour. *Nature* 645, 177-191.

Findling C, Hubert F, International Brain Laboratory (2025) Brain-wide representations of prior information in mouse decision-making. *Nature* 645, 192-200.

Google Scholar Page:

https://scholar.google.pt/citations?user=ivDH8SoAAAAJ